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Abstract

This study sought to establish the most important factors affecting the service in high-level women's volleyball and the relative weighting of such factors on this technical part of the sport. A total of 1300 services from eight matches played in two Final Fours of the Indesit European Champions League were analysed. The services were delivered by 58 players of 25 nationalities. Observation sheets and two video cameras located at both ends of the court were used. Service speed was measured by radar. The twelve variables studied enabled the service to be divided into four components. The most influential component (19.02% of total variance) comprised variables related to technical service characteristics (type of service and service speed). The second most influential component (15.16% of variance) was related to the opponents' technique and tactics, and to their position on court at the time of the service. The service was also affected by the technical and tactical movements that the servicer needed to perform in the subsequent play (12.20%). The stage of the match and the score (10.67%) also presented players with different levels of risk and helped to determine the type of service chosen and the power with which it was executed.

KEYWORDS: volleyball, skill, service, principal component analysis

Introduction

The International Volleyball Federation (FIVB) has introduced a number of important rule changes over the last 15 years in an effort to make the sport more attractive as a spectacle. These rule changes have recently been studied by a number of authors (Ejem, 2001; Lozano, 2007; Lozano, Calvo, Cervelló & Ureña, 2003; Martínez & Abreu, 2003; Morales, 2000;Palao, Santos & Ureña, 2004; Quiroga, García-Manso, Bautista & Moreno, 2008). Some changes have tended to reinforce the importance of the service, thus making serving skill a key contributor to final victory. The following are among the major changes introduced: a) Extension of the service zone to the nine metre area behind the service line – the servicer does not commit a positional fault (introduced in 1994); b) Elimination of the "double hit" fault for the team's first touch (introduced in 1994); c) Introduction of the libero player (1998); d) Introduction of the Rally Point System in each of the sets played (1999); e) Elimination of the attempted service (introduced in 1999); f) Increase in the time allowed for serving to eight seconds (introduced in 1999); and g) Allowing the service ball to touch the upper edge of the net on passing to the opponents' court (introduced in 2000).

For practical purposes, these changes can be classified into three groups: those that affect scoring (d), those that affect the composition of the teams (c) and those that directly affect the service (a, b, e, f, and g).

The service is the only moment of the game when the player has 100% control over the ball and the way events unfold; she needs to choose the type of service, the force she will impart to the ball and the area of her opponents' court to which the service will be directed (Martínez & Abreu, 2003; Ureña, 1998). In other words, underlying the service is the opportunity to create the maximum number of problems for the receiving opponent and, if possible, score a direct point (Ureña, 1998; Ureña, Santos, Martínez, Calvo & Oña, 2000).

As a result, the service has become the team's preeminent offensive weapon; it is capable of causing a range of problems for the receiving team, restricting, forestalling and even eliminating the possibility of an effective attack on the part of the opponents (Díaz, 2000; Katsikadelli, 1998b; Lozano, 2007; Molina, Santos, Barriopedro & Delgado, 2004; Papadimitriou et al., 2004; Tsivika & Papadopoulou, 2008; Ureña, 1998, 2008; Ureña, Calvo & Gallardo, 2000).

The overall performance of a volleyball team depends on a chain of multiple factors. These factors are decisive for the deployment of the technical and tactical skills that lead directly to the winning or losing of a play or rally. Outstanding among these are failed services and aces (Drikos, Kontouris, Laios & Laios, 2009; Marcelino, Mesquita & Afonso, 2005; Marelic, Resetar & Jankovic, 2004).

Marcelino et al. (2005) argue that the execution of failed services and the points deriving from services are decisive aspects of high-level volleyball. In the same vein, Palao (2001), at the 2000 Sydney Olympics, and Valadés (2005), at the 2004 Athens Olympics, report that the service accounts for between 4.4 and 8.1% of the points obtained by the various final actions.

Consequently, the service has become one of the main components in a team's play, and is a key element for the trainer to consider when drawing up training programmes. One of the most important effects of the rule changes has been a change in the types of service most frequently used in international volleyball. The increased use and current predominance of the jump service is notable in high-level men's volleyball (Agelonidis, 2004; Callejón, 2006;; Ejem, 2001; Katsikadelli, 1998a; Tsivika & Papadopoulou, 2008), while in women's volleyball this type of service has developed more slowly (Ejem, 2001; Quiroga et al., 2008; Ureña, 1998). This difference may be due to gender-related differences in build, as well as to the risks inherent in executing this type of service.

This study sought to determine the factors of the game and its progress that most affect the service in high-level women's volleyball, and the relative weighting of these factors over the course of a game.

Methodology

Sample. The sample consisted of 1300 services drawn from the 29 sets of eight matches played in the Final Four of the *Indesit European Champions League*, held in Santa Cruz de Tenerife (Spain). These services were delivered by a total of 58 players of 25 different nationalities, playing for the four best teams in the European Club League in the two championships studied. The sample includes 52 players who had represented their countries, of whom 16 had taken part in the Olympic Games. Only six players had not previously played in their national teams.

Observation criteria. Observations were made using a recording obtained from two Panasonic NVDS88 digital cameras, placed at either end of the court, with the aim of obtaining an optimal field of view for the variables selected.

A hand-held radar was used to measure service speed (Stalker Professional Radar, Deluxe Stats Package; Plymouth MN, USA), as validated by the research of De-Renne Ho and Blitzblau (1990), Kraemer et al. (2000) and Valadés, Palao, Fernia, Padial and Ureña (2007). The radar operator recorded the speeds from the back of the court, opposite the position from which the service was delivered, making appropriate positional changes for each service in order to align the radar with the ball's trajectory, as suggested by Ferris et al (1995), and Forthomme, Croisier, Ciccarone, Crielaard and Cloes (2005), to minimise possible measurement errors.

To ensure the reliability of observations, the degree of intra- and inter-observer agreement was checked, using Bellack's formula (Van Der Mars, 1989), with a 10-day time interval. 20% of the sample was checked, (services drawn from four matches in the sample, chosen at random). The reliability scores thus obtained were greater than 95% for all the variables under investigation.

Data analysis. The importance of the service, and its role in determining victory, was initially assessed in terms of twelve previously-chosen variables. Variables were selected as the result of an earlier discussion between three expert traineres with working experience of professional volleyball teams. After analysing the chosen variables in control matches, two were eliminated when they were found to overlap with other variables under investigation.

The variables finally used were as follows: 1) the type of service (power jump service, jump float service, float tennis service, power tennis service, float hook service); 2) the speed of the service (<40 km/h, 40-49 km/h, 50-59 km/h, 60-69 km/h, 70-79 km/h, 80-89 km/h and >89 km/h); 3) the zone at which the service is aimed (six zones, three forward and three rear) following the methodology suggested by Wise (2002) and Mesquita, Manso and Palao (2007); 4) the playing position of the receiving player (outside, opposite, middle or libero); 5) the efficacy of the service, using the categories set out in the FIVB system, adapted from Coleman (1975) (ace; service that makes mounting an attack impossible and leads to a free ball; service that limits attacking options, forestalling rapid attacks; service allowing any type of attack to be mounted; and failed service); 6) zone from which the service is delivered (three zones, identified as the extension of defensive zones 1, 5 and 6) and chosen according to the methodology put forward by Gerbrands and Murphy (1995); 7) playing position of the servicer (setter, outside, middle or opposite player); 8) score at the time; 9) score bracket at the time of the service (for the first four sets the following brackets were used: 1-8; 9-16; 17-20 and 21-25; while two margins were used for the fifth set: 0- 8 and 9-15); 10) number of set (first, second, third, fourth or fifth set).

Data for the final variables were subjected to Principal Component Analysis, with a view to synthesising the information, reducing the number of variables and grouping together the most important variables in the volleyball service without losing the weightings provided by the original information. The respective weightings of each of the factors (components) obtained from the matches under investigation involving high-level women players were calculated in this manner.

Results

Table 1 shows the proportion of variance accounted for by the four main factors identified by Principal Component Analysis, together with the correlations between each factor and the variables analysed.

It will be noted that the variables grouped themselves around specific aspects of play related to: the execution of the service (factor 1), the receiving team (factor 2), the servicer (factor 3) and the state of the game (factor 4). Factor 1 accounted for 19.02 % of total variance in the service, while factor 2 accounted for 15.16 %, factor 3 for 12.20 % and factor 4 for 10.67 %.

Variable	*Factor 1 (19.02 %)*	*Factor 2 (15.16 %)*	*Factor 3 (12.20 %)*	*Factor 4 (10.67 %)*
Playing position of player executing service	0.2981	-0.1198	**0.6438**	-0.0034
Type of service	**-0.6476**	-0.0316	0.0147	-0.0508
Speed of service	**0.6484**	0.0158	-0.0321	0.0304
Zone in which service is executed	-0.1873	-0.2063	**-0.7071**	-0.0541
Zone at which service is aimed	-0.0680	**-0.6485**	-0.1024	-0.0007
Playing position of receiving player	0.0907	**-0.5277**	-0.0836	0.2349
Efficacy of service	0.0414	**-0.4853**	-0.2088	-0.2394
Score	0.0353	-0.0655	-0.0986	**0.6509**
Stage of game	-0.1062	0.0498	0.0724	**0.5445**
Set	-0.0956	0.0229	0.0910	**0.4011**

Table 1. Proportion of variance of the four factors identified by Principal Component Analysis, and the correlations of each factor with the analysed variables.

Discussion

Analysis of variables for the 1300 services studied enabled the service to be divided into four components whose total weighting, within the overall result, accounted for 57% of the explained variance. It should be stressed that, for the principal component analysis, all the variables used were considered important for the execution of an effective volleyball service. Each of these components includes variables that differed with respect to one another, which does not apply to the other components influencing the service.

The component with the greatest influence (19.02% of the variance) was related to the type of service used (factor 1), highlighting the importance of the technical execution and the characteristics of the chosen action, as well as the speed imparted to the service ball. The players try to execute a service that secures a direct point or that at least impedes the opposing team from mounting an attack, and to this end they aim for a service that is accurate, difficult to receive and very often powerful.

The power jump service, which represented 23.92% of all the services made, is the most decisive service in high-level women's volleyball, although it is not the most frequently used by the players covered by this study (overhead float service: 48.62%). It is important to bear in mind that the type of service used (trajectory and speed of the ball) is always going to influence its reception and the tactical response of the defending team (Ureña et al., 2001).

A determining factor in the choice of service was the servicer's playing position in the team. Most services by opposite players had a speed of >80 km/h (31.0%). Those by outside players frequently reached 70-79 km/h (13.3%) and services by middle players commonly achieved speeds of 50-69 km/h (65.0%). The lowest speeds (<49 km/h, 23.0%) were achieved by setters.

The results clearly revealed the tactical intentions of setters in the use of the overhead float service, as well as the priority opposite players gave to strength in their use of the jump spin power service. This concurs with findings by Frönher (1995), Ejem (2001) and Katsikadelli (1996) for male volleyball players. In the present study these choices were deemed to be largely the result of the physique and fitness levels of players.

Both in men's and women's volleyball, the jump spin power service is the service that prevents an opposing attack the highest percentage of times (Palao et al., 2004; Quiroga, García-Manso, Moreno & Bautista, 2007). Morales (2000) suggests that the number of players per team who execute jump spin power services, especially in the men's game at the elite level, has become ever higher and the extra speed of this service forces an increase in the number of players participating in its reception. To the best of the present authors' knowledge, this is the first study to assess the most important playing variables affecting the service in women's volleyball in competitive conditions at an elite level – in this case the Final Four of the *Indesit European Champions League,* and with a statistically significant sample (1,300 services).

It should also be stressed that the speed of the ball and the effectiveness of the service will rise as the level of the player rises or as her specialisation in serving increases. This is to be expected, given the risks run by a player who chooses a jump power service compared to another who opts for a more tactical service (e.g. jump float) in which the ball is directed towards the worst receivers, towards overwhelming a single receiver, or towards the weakest zones.

The jump spin power service, mostly used by opposite and outside players, was the fastest type of delivery, as reported by Ejem (2001) and Katsikadelli (1996). In the present study, the 10 fastest services recorded were all made by one opposite player. The mean service speed was 100.6 km/h, and the fastest was 107 km/h, slightly faster than the 96 km/h maximum recorded by Ejem (2001) at the Sydney Games. Uriarte (2007), trainer of the Argentinian men's team, recently stated that delivery speeds of >95 km/h pose a major problem for receivers (outside players and liberos). In women's volleyball, speeds slightly lower than this can be difficult for receivers to handle, but in this study it was observed that only opposite players took significant advantage of this by delivering very powerful services. In tactical terms, these are the players who are allowed to take greater risks when serving.

The speed of the service ball is influenced by the technical characteristics of the service (type of service). The average speed of the jump power services was 78.9 km/h (+/- 9.37), compared with 54.9 km/h (+/- 5.09) for the jump float service, 54.4 km/h (+/- 5,89) for the tennis float service, 54.1 km/h (+/- 6.06) for the power tennis service, and 51.9 km/h (+7- 4.51) for the hook service.

The next most important aspect of the service, after the choice of service type, included variables relating to the receiving team (zone at which the service is aimed, position of the receiving player in the team and the effectiveness of the service) – in other words the receiving player's profile and the effect the chosen service has on the opposing team's system of attack. These variables accounted for 15.16 % of observed variance. These findings reflect an important tendency in high-level volleyball: deliberately aiming the service towards particular zones and players with the intention of increasing the effectiveness of the service, hampering reception and, therefore, the tactical system of change of service (reception, positioning and attack).

Although several varieties of service and attack exist, the types most commonly used by elite-level teams show certain similarities, particularly in the phases of armswing and ball contact (Rokito, Jobe, Pink, Perry & Brault., 1998), and jump and flight. It should be noted that, when executing a jump service or spike, the player commonly hits the ball at the highest point of ascent in order to send the ball to the other side of the court at the highest possible speed (Tillman, Hass, Brunt & Bennet 2004). Therefore, the higher the level of the player, the faster the ball will travel and the more effective the hit will be. Moras et al. (2007), in their study of elite men's volleyball players, reported similar ball speeds attained in both serving and attacking (97-100 km/h) (Coleman, Benham, & Northcott, 1993; Forthomme et al., 2005).

The third most important factor in terms of the variables influencing the service in high-level women's volleyball relates to the individual characteristics that vary with the profile of the servicer. The choice of service zone is determined by the position that the servicer habitually occupies and by the subsequent play she wishes to perform. It is important to bear in mind that after the execution of the service the player has to cover a specific area of the court that allows her to fulfil her subsequent defensive duties. This factor accounted for 12.20 % of the variance between the services aspects analysed.

The service area most frequently used was behind zone 1, as reported by Maia and Mesquita (2006). This was seen for all playing positions except middle players, who preferred to service from behind zone 5. The players studied tended to service from behind the zone they subsequently occupied during defence: setters and opposite players from behind zone 1 (100% and 80%, respectively), which they subsequently defended, and middle players from behind zone 5 (47%), which they similarly defended after serving. These results can be explained by the players' wish to facilitate transition from the service area to the defence zone by reducing the amount of ground they need to cover, while complying with the play system devised (Gerbrands & Murphy, 1995; Frönher, 1997; Maia & Mesquita, 2006; Mesquita et al, 2007). In the case of outside players, 56.2% serviced from behind zone 1 and defended in zone 6, 18.8% serviced from behind zone 6 and defended in zone 6, 18.8% serviced from behind zone 5 and defended in zone 6, and 6.2% serviced from zone 5 and defended in zone 5. This strategy is a response to the technical aspects of the service and the type of service used.

The way the match is developing (factor 4 - state of the game) encompasses the last group of variables (score, score bracket and set number) that must be taken into account when executing one type of service or another. The relative weight of this factor was found to be 10.67 %. The findings indicate that these variables are only important in specific situations in the game and do not have the same importance for the entire duration of a set.

García-Tormo, Redondo, Valladares and Morante (2006), who studied the services of young Spanish female players, reported a small but significant connection between the level of risk the players run and the points difference obtaining at the time of service. Molina and Barriopedro (2003) report that, in the national men's first division, service type changes in accordance with the score in such a way that, as the difference in score fell, so too did the use of the jump service, while the greater the winning margin the greater the use of this more risky type of service.

Conclusion – Practical Applications

The main variables influencing the service in high-level women's volleyball are those related to the technical characteristics of the execution (type and speed of service). Secondly, the service is dependent on the technical and tactical characteristics of the opposing players and the positions that these players take up on the court at the time the service is delivered. The service is also affected by the technical and tactical movements that the servicer needs to make in subsequent play. Finally the stage of the match and the score need to be considered, since these variables represent different levels of risk for the servicers and help to determine the type and power of the service that is executed.

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